

MODEL QUESTION PAPER
TED(15)-4262
FOURTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY
DATA COMMUNICATION AND COMPUTER NETWORKS

(Maximum marks: 100)

(Time: 3hrs)

PART-A

I [Answer the following questions in one or two sentences. Each carries '2' marks]

1. What is a protocol ?
2. What is meant by Physical address ?
3. What is routing ?
4. Define Port address ?
5. How many classes are there in IPv4 addressing scheme ? (5 x 2=10)

PART-B

II [Answer any five of the following questions. Each question carries '6' mark]

1. Describe the difference between Analog and Digital data ?
2. Write short note on classifications of computer networks based on geography ?
3. Write notes on various transmission mediums used for guided transmission ?
4. What is meant by topology ? Give examples for different topologies ?
5. Describe briefly about CSMA/CD ?
6. Explain ethernet frame format with a diagram.
7. What is the difference between error detection and error correction ? (5x6=30)

PART-C

[Answer one full question from each unit. Each question carries '15' mark]

Module 1

- III a) What is the difference between analog and digital data ? (5)

b) Write notes on Transmission impairments that occurs in digital transmission ? (10)

OR

IV a) Differentiate between guided and unguided transmission ? (15)

Module 2

V a) Compare ISO-OSI architecture and TCP/IP architecture ? (15)

OR

VI a) Explain the schemes FDMA, TDMA and CDMA ? (15)

Module 3

VII a) What is IPv4 ? (3)

b) Explain classfull addressing scheme of IPv4 ? (12)

OR

VIII a) What is meant by datagram ? (3)

b) Explain the UDP datagram format ? (12)

Module 4

IX a) What is an Email ? (3)

b) Explain the working of email system with necessary diagrams. (12)

OR

X a) What is FTP ? (3)

b) Describe the working of FTP ? (12)